
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

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Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ -VLA, a rational vision-language-action model family with a 19.2M convolutional policy and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned operations. Their weights can therefore be reorganized into explicit interaction tensors rather than treated only as an opaque checkpoint. A runtime operator audit of the seed-0 20.1M ViT checkpoint and local numerical reconstructions verify its operator-level form. Across three independently trained 20.1M ViT policies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint. A fixed elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$ at the cost of three forward passes. A separate convolutional instantiation passes an independent 4,608-case joint-attention certificate with worst relative error 6.36×10^{-16} . Exact-attention records naming those same ViT checkpoint artifacts numerically replay every attention module on one fixed corrected cached input per checkpoint, with worst relative error 2.016119×10^{-7} . In a seed-0 ViT checkpoint, exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. Separately, prospectively fixed rank-96 visual projectors retain 95.2–98.2% of baseline successes on tasks absent from corrected projector construction, while three equal-rank random projectors retain only 3.3–13.6%. The result is a capable and structurally inspectable specialist family with exact access to trained attention coefficients. It does not yet provide an exact end-to-end decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

This is not a zero-shot policy. It is trained from scratch on one specialist suite. Its role in the workshop’s question is to establish an auditable specialist control against which future zero-shot and transfer claims can be evaluated.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses

additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on 1,500 canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

79 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

80 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
81 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
82 interaction coefficients determined by the weights.

83 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
84 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
85 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

86 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
87 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
88 rescaling without leaving the rational tensor representation. Training and deployment use r itself. The
89 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
90 not uniform agreement with RMSNorm. Appendix A.3 quantifies that distinction.

91 2.2 Architecture and training

92 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
93 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
94 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both
95 share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-
96 query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
97 blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator
98 census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three
99 stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional
100 files receive the joint-attention and rational-normalizer certificates reported in Appendix A.3.

101 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
102 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
103 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
104 that verify every requested state was loaded.

105 2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

106 Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M
107 ViT checkpoint at runtime and reconstruct representative operations from saved weights.

108 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
109 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
110 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
111 summarized in Appendix Figure 4. It does not materialize one compact polynomial for the full
112 eight-layer transformer.

113 3 Protocol-aligned closed-loop capability

114 3.1 Evaluation protocol

115 We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 116 • 50 canonical trials per task
- 117 • a 280-step cap
- 118 • three independently trained ViT checkpoints
- 119 • 500 rollouts per checkpoint and 1,500 rollouts in total

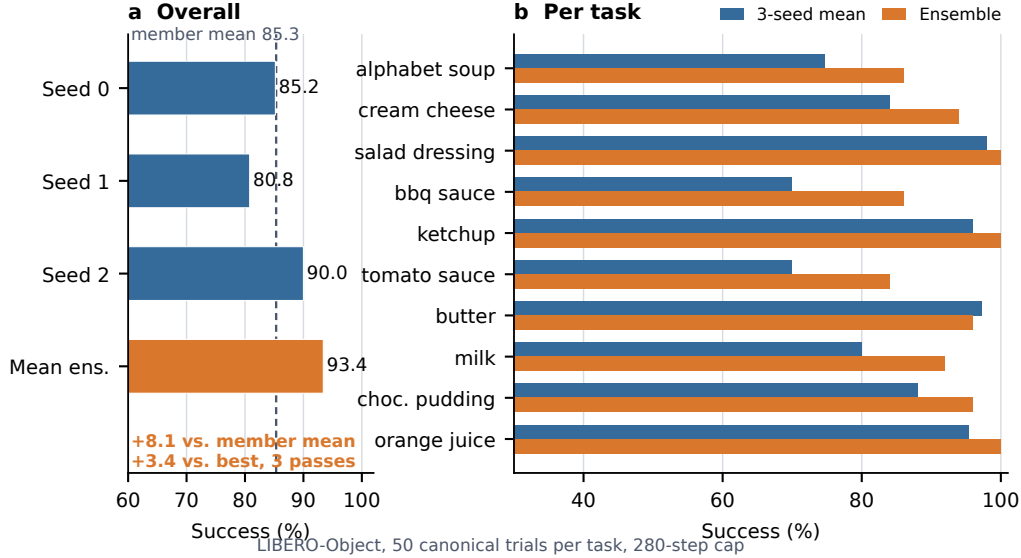


Figure 1: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ -VLA, rational ViT	20.1M	1 pass	85.3% $\pm$ 4.6%
χ -VLA, prediction-mean ensemble	$3 \times 20.1M$	3 passes	93.4%

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates

137 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
 138 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
 139 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

140 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 141 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 142 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 143 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 144 10^{-16} . On one fixed corrected cached input per checkpoint, we numerically replay all four vision
 145 and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts
 146 as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst
 147 relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5} . This residual
 148 comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself
 149 uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint
 150 records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

151 We separately certify three convolutional checkpoints from the same rational architecture family.
 152 Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all
 153 twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs
 154 per checkpoint yield 384 module cases and 4,608 head-input cases. Every output is finite and the
 155 worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This same-checkpoint
 156 certificate covers the joint-attention equation at each normalized layer input. It does not rebuild the
 157 convolutional stem, feed-forward blocks, residual composition, final normalization, or action head as
 158 one end-to-end contraction.

159 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 160 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
 161 rank important directions without constructing the full table. It matches brute-force materialization to
 162 2.13×10^{-14} at toy scale.

163 4.2 Trained policy result

164 The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking
 165 experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived
 166 calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random
 167 projector banks are sampled once, then fixed and reused for all examples. The weights choose the
 168 subspace, while cached activations are used only afterward to score how well it preserves the layer’s
 169 computation.

170 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 171 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 172 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 173 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent
 174 model-level replicates.

175 This extends exact weight-only construction through individual attention layers. It is not an exact
 176 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 177 rapid degree and representation growth.

178 A separate intervention tests robustness at the visual-to-joint interface rather than deriving a projector
 179 from the exact attention factors. An action-Jacobian projector estimated on official Object tasks
 180 0–3, with rank 96 fixed before corrected outcomes, retains 259/272 full-policy successes on tasks
 181 8–9 across three ViT checkpoints. These tasks are absent from corrected projector construction.
 182 An activation-energy projector retains 267/272, while three equal-rank random projectors retain
 183 37/272, 9/272, and 22/272. All ten tasks appeared during policy training. This shows robustness to a
 184 structured 75% reduction of the internal channel space, not zero-shot transfer, unique action-Jacobian
 185 specificity, or a causal link to the exact weight factors. Appendix A.2 reports the paired controls and
 186 stability intervals.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The ViT records connect capability, exact attention, and the reduced-Gram screen through common checkpoint basenames. Separate convolutional files replicate the joint-attention equation certificate across another visual encoder. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The anonymous artifact releases raw structural-certificate JSONs and all 2,000 episode rows behind the ViT member and ensemble results. Immutable SHA-256 manifests and standard-library verifiers independently recompute the capability totals, visual-bottleneck counts and intervals, and the convolutional-attention and RationalNorm certificate decisions. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. Canonical-state checks fail loudly when requested states are unavailable.

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A Supporting results and decisions

A.1 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.2 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

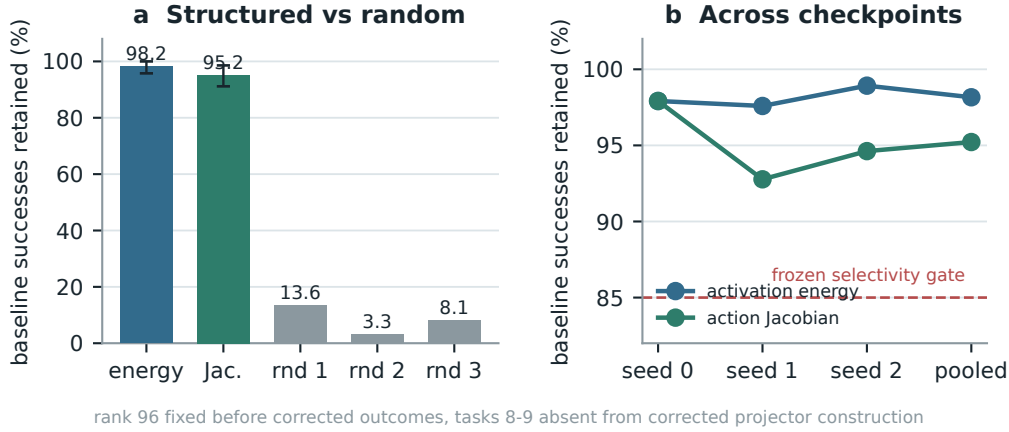


Figure 2: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This is not zero-shot evidence, and the data-derived intervention remains separate from the exact weight-only attention decomposition.

A.3 Numerical scope of rational conversion

The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients $(5.32997, 7.53552, 0.327274)$ and the denominator has coefficients $(1, 9.64827, 2.60664)$. Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate.

The certificate also bounds what the operator approximates. Observed normalized mean-square ratios v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs, the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

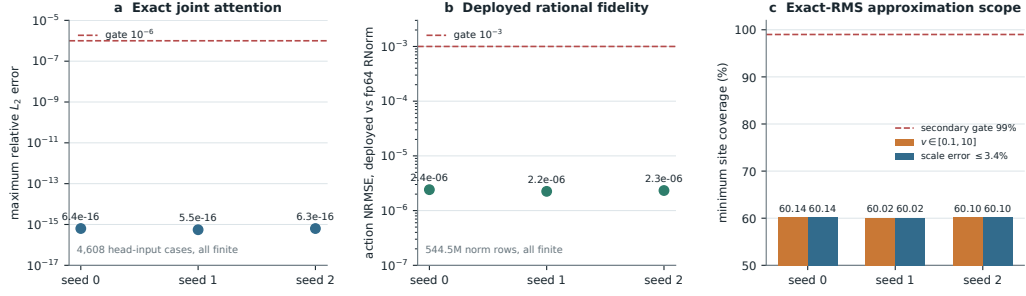


Figure 3: **Frozen structural certificates on three independent convolutional checkpoints.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.

Mechanical operator audit of the seed-0 20.1M ViT checkpoint

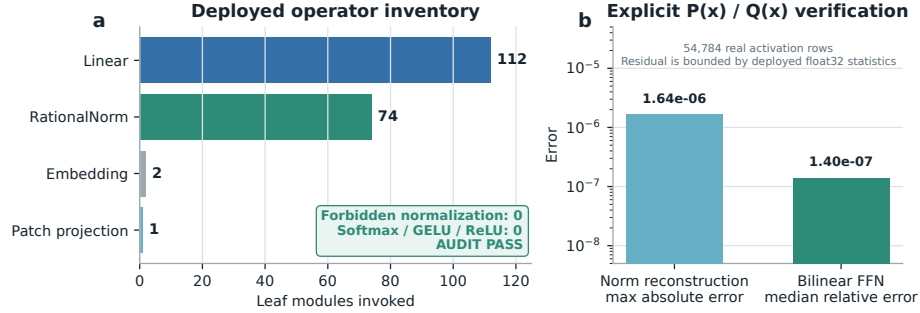


Figure 4: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.